



Figure 1: Ubuntu OS installed in a VM

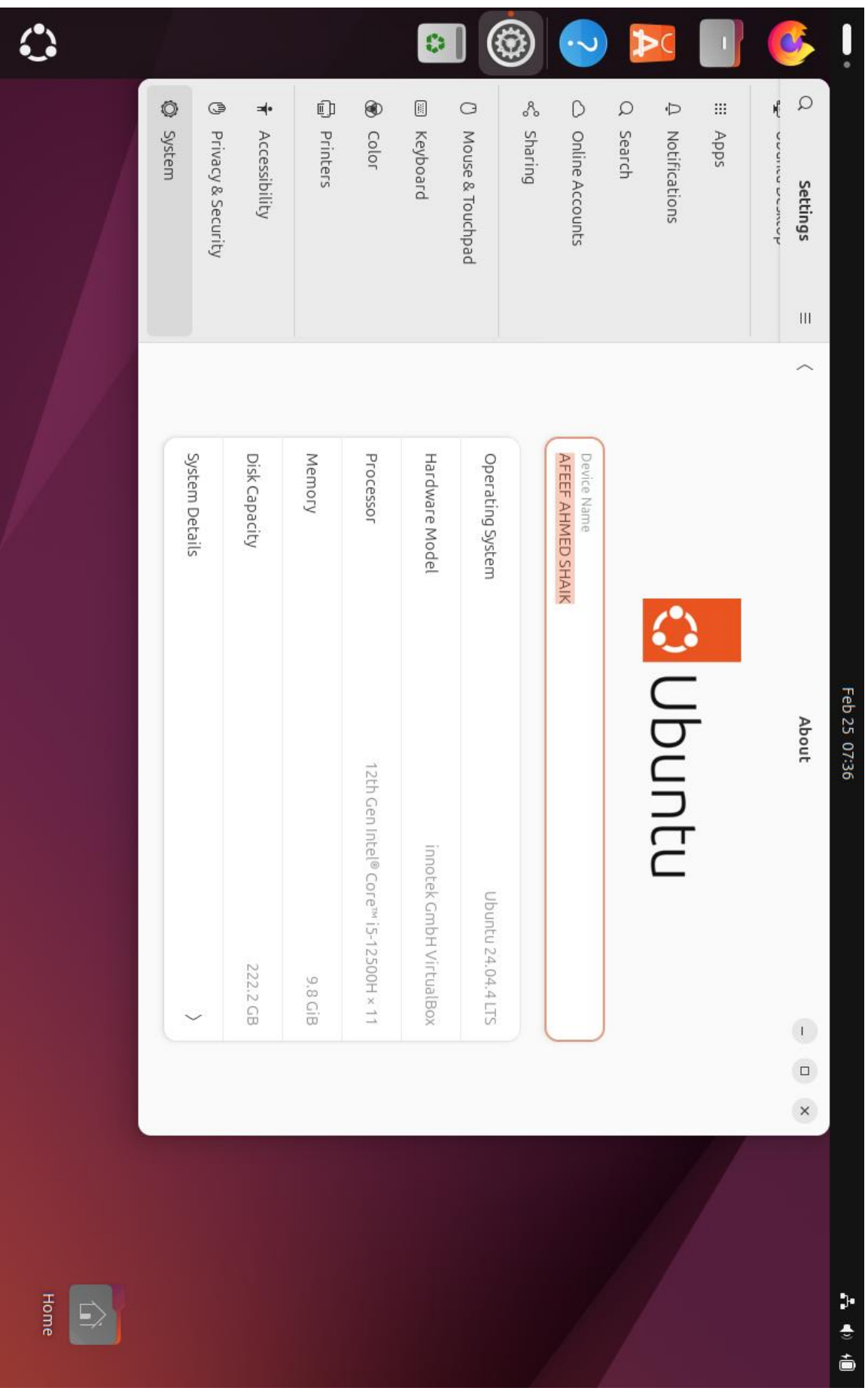


Figure 2: USER PROFILE

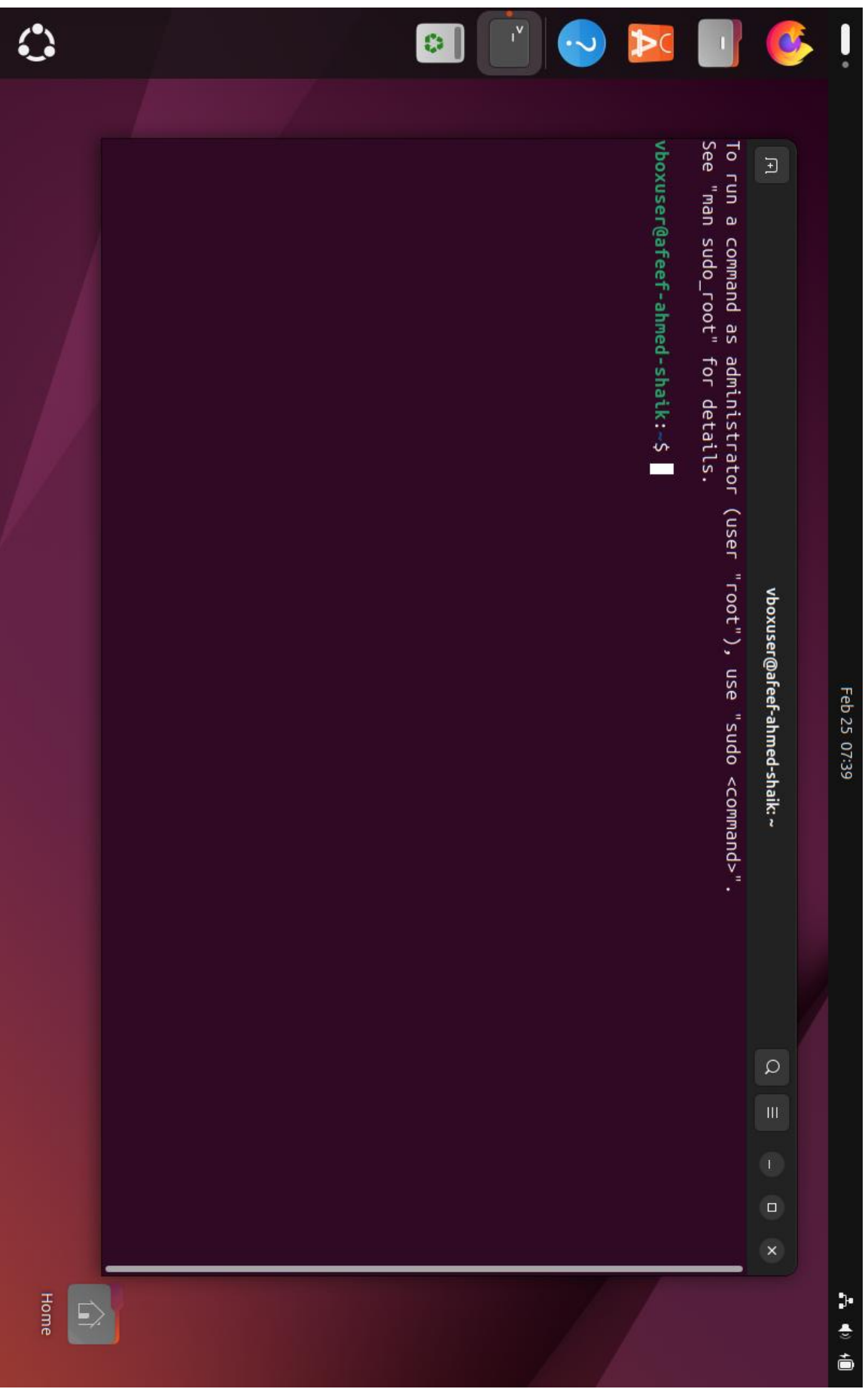


Figure 3: USER TERMINAL

NGC Catalog

All you need to build AI—GPU-optimized containers, pretrained models, SDKs, and Helm charts—unified in one catalog for cloud, data-center, or edge.

Top Technology

The most-downloaded AI frameworks, models, and SDKs on NGC—curated by NVIDIA, battle-tested by the community, ready to accelerate your next project.

- All
-  Containers
-  Collections
-  Models
-  Resources
-  Helm Charts

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 **NVIDIA PyTorch**

PyTorch is a GPU accelerated tensor computational framework. Functionality can be extended with common...

High Performance Comput...

Natural Language Understand...

 Container Updated 22 days ago

 **NVIDIA Triton Inference Server**

Triton Inference Server is an open source software that lets teams deploy trained AI models from any framework, from...

Object Detection

Automatic Speech Recognition

 Container Updated 20 days ago

 **NVIDIA cuda**

Container registry for CUDA images

CUDA

 Container Updated a month ago

 **NVIDIA DeepSeek-R1**

NVIDIA NIM for GPU accelerated DeepSeek-R1 inference through OpenAI compatible APIs

 +1

 **NVIDIA Cosmos World Foundation Models**

Cosmos World Foundation Models: A family of highly performant pre-trained world foundation models purpose...

 +1

 **NVIDIA DeepStream SDK**

NVIDIA DeepStream SDK enables developers to build accelerated pipelines for a wide range of use-cases such as...

 +1

Figure 4: NVIDIA NGC USER DASHBOARD

Organization

Dashboard

Audit

Organization Profile

Teams

Usage

Activate Subscription

Subscriptions

Webhook Management

Setup > API Keys

API Keys

Personal Keys

Search table

Key ID

Key Name

Expiration

Key Value

Status

87357941-2298-4b21-ab36-356dde07fcac

AFFEEF AHMED SHAIK

02/25/2027

nvapi-*****g5q

ACTIVE

25 > Items

<< < 1 > >>

Go to 1

NGC API Keys

NVIDIA NGC API keys are required to authenticate to NGC services using NCG CLI, Docker CLI, or API communication. NVIDIA NGC supports two types of API keys.

Personal API Key

Any user who is a member of an NGC.org can generate Personal API Keys. These keys are tied to the user's lifecycle within the NGC.org and can access up to the permissions and services assigned to the user. During the key generation steps, users can configure which NGC services are accessible by the API key and the time-to-live from one hour to 'never expires'.

Legacy API Key

This is the original type of API key available in NGC since its inception. This type allows you to create only one 'API key' at a time. Generating a new key automatically revokes the previous one, as they cannot be rotated. The active key immediately becomes invalid when you create a new key.

7 AFFEEF AHMED SHAIK
AFFEEF NVIDIA CLOUD

+ Generate Personal Key

Figure 3: USER TERMINAL

Organization

Organization

Dashboard

Audit

Organization Profile

Teams

Usage

Activate Subscription

Subscriptions

Webhook Management

Setup > API Keys

API Keys

Personal Keys

Search table

Key ID
87357941-2298-4b21-ab3f
be3-c91a-625d-4906-9cdf

25 Items

Generate Personal Key

The following Personal API Key has been successfully generated for use with this organization. This is the only time your key will be displayed.

\$ nwapl-ngpD3uk8sxEFcM4ESGJ92b1lw8Dnk-e8Hcbj088NkQZwI5to4 H0ZJYz2GH_TMDD

Keep your Personal Key a secret. Do not share it or store it in a place where others can see or copy it.

CloseCopy Personal Key

NGC API Keys

NVIDIA NGC API keys are required to authenticate to NGC services using NGC CLI, Docker CLI, or API communication. NVIDIA NGC supports two types of API keys.

Personal API Key

Any user who is a member of an NGC org can generate Personal API Keys. These keys are tied to the user's lifecycle within the NGC org and can access up to the permissions and services assigned to the user. During the key generation steps, users can configure which NGC services are accessible by the API key and the time-to-live from one hour to never expires.

Legacy API Key

2 Results

View

Status

ACTIVE

ACTIVE

Go to 1

Figure 6: NVIDIA NGC
API KEY GENERATION

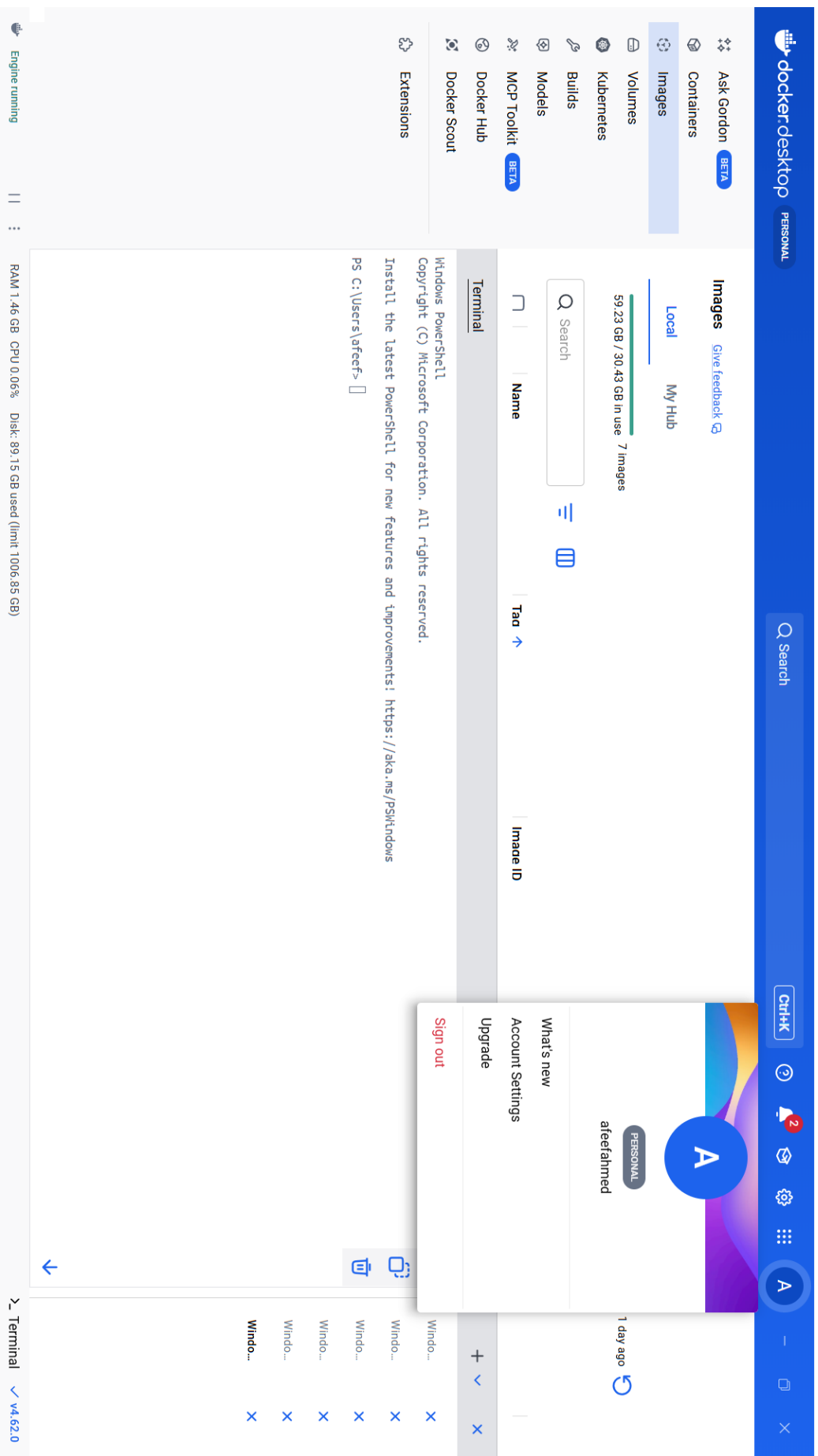


Figure 7: Docker Images Before Pulling

dockerdesktop

PERSONAL

Q Search

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🔄 Images

💾 Volumes

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📦 Models

🔧 MCP Toolkit

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🔍 Docker Scout

🔌 Extensions

Images

Give feedback

Local

My Hub

42.12 GB / 30.43 GB in use

3 images

Q Search

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☐	Name	Tag ↑	Image ID	Created	Size	Actions
☐	nvcrl.io/nvidia/tensorflow	25.01-tf2-py3	4b15081766f8	1 year ago	31.97 GB	🚀 🔍 🗑️
☐	nvcrl.io/nvidia/pytorch	26.01-py3	38ed2ecb2c16	1 month ago	30.43 GB	🚀 🔍 🗑️

Last refresh: 2 hours ago

🔄

Terminal

Error response from daemon: failed to resolve reference "nvcrl.io/nvidia/tensorflow:26.01-tf2-py3": nvcrl.io/nvidia/tensorflow:26.01-tf2-py3: not found
PS C:\Users\afref> docker pull nvcrl.io/nvidia/tensorflow:25.01-tf2-py3
25.01-tf2-py3: Pulling from nvidia/tensorflow
4f159b21c0f: Pulling fs layer
e852726cdf4b: Pulling fs layer
a120ee82007: Pulling fs layer
7dfa9293b99: Pulling fs layer
8135582f4ed: Pulling fs layer
07e412f82f03: Pulling fs layer
dc44b265507a: Pulling fs layer
8ec2f291c690: Pulling fs layer
2dabfd5ddd1: Pulling fs layer
4eb34e451322: Pulling fs layer
75f58769314d: Pulling fs layer
4f4fb70ef54: Pulling fs layer
5a9563dd6b7e: Pulling fs layer
4459897c0dc0: Pulling fs layer
e5a680752b35: Pulling fs layer

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Windo...

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Engine running

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RAM 7.29 GB CPU 0.00% Disk: 76.65 GB used (limit 1006.85 GB)

> Terminal

✓ v4.62.0

Figure 8: TensorFlow Container Pull

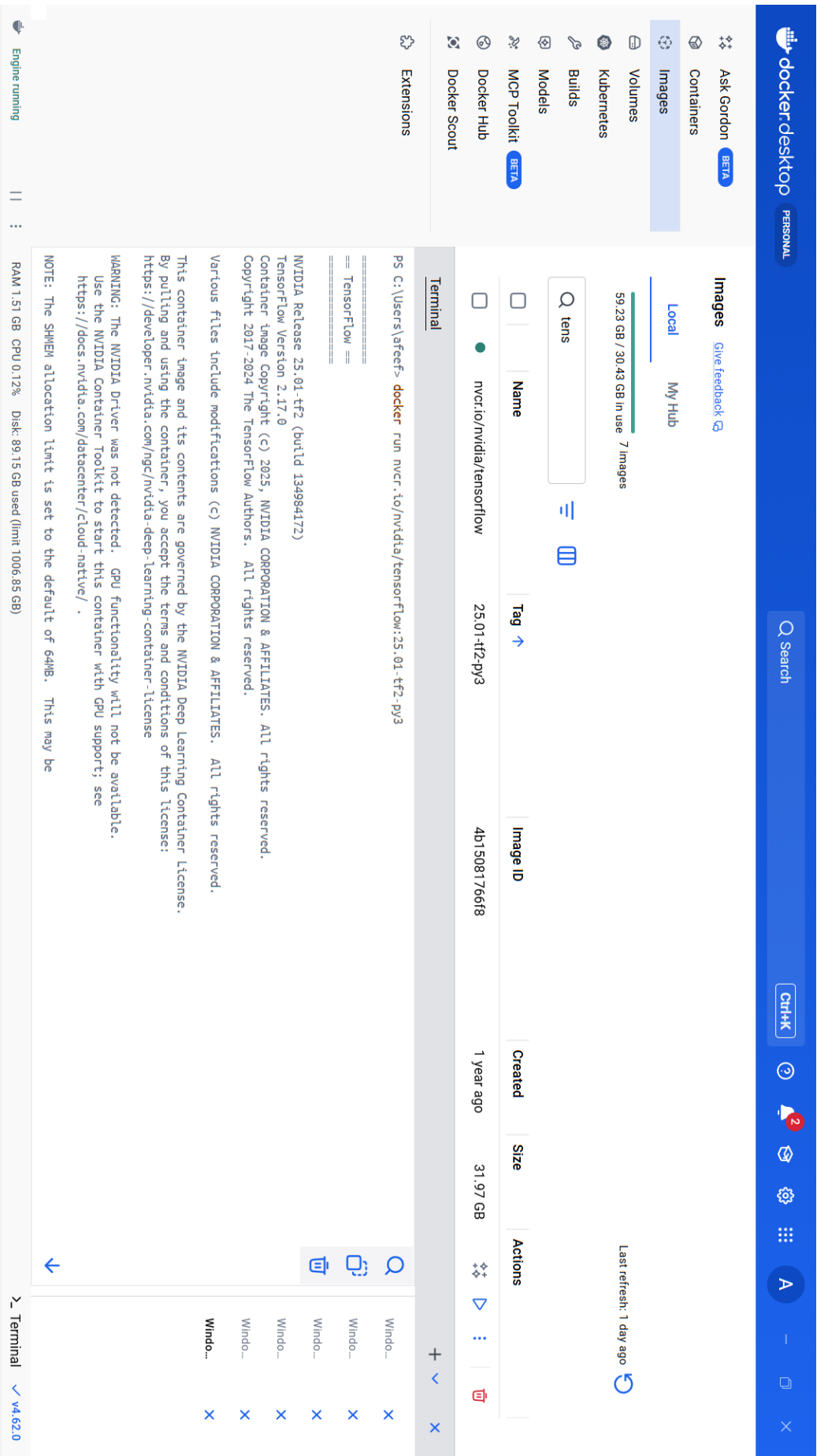


Figure 9: TensorFlow Container Run

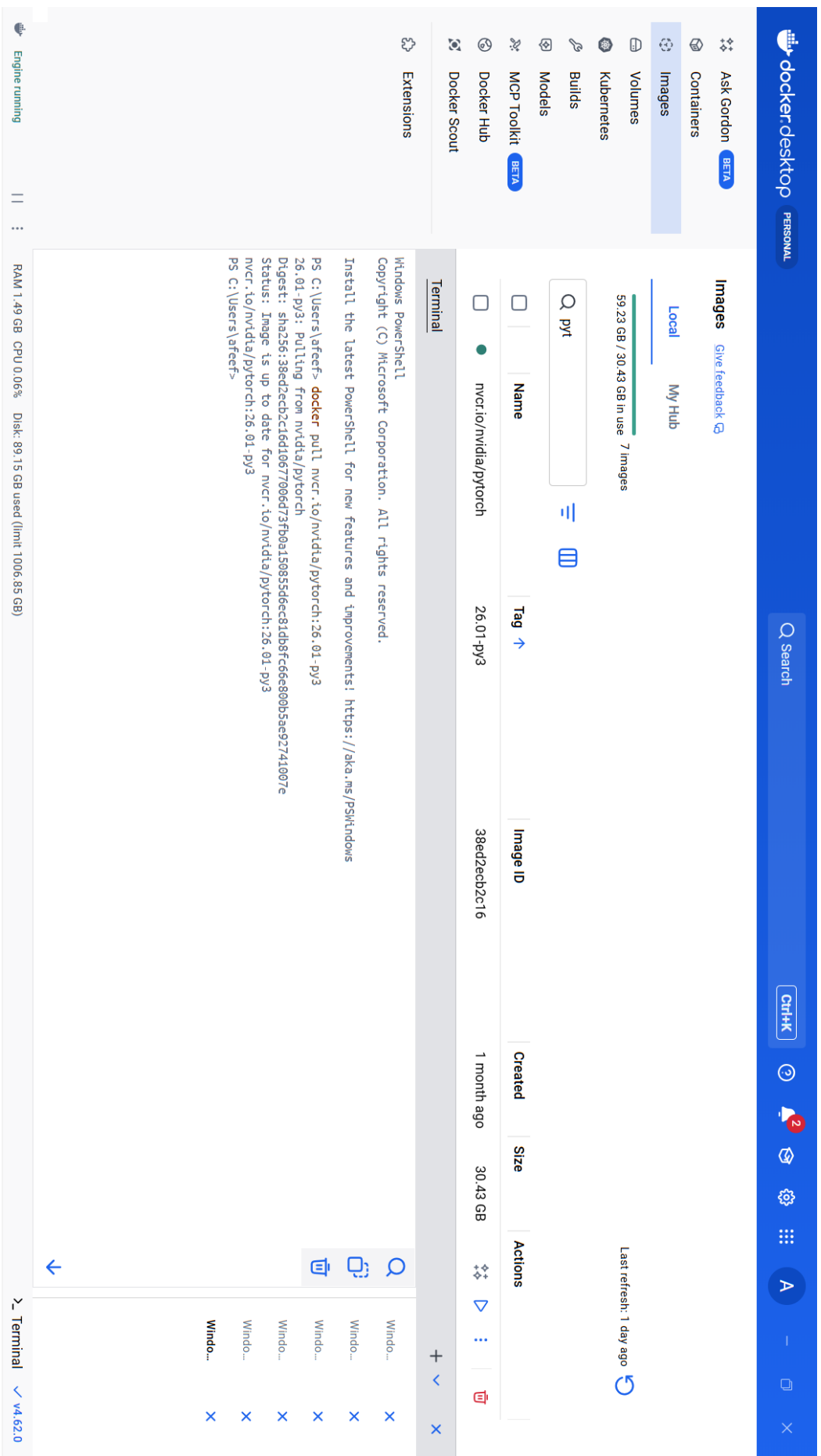


Figure 10: PyTorch Pull

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dockerdesktop

PERSONAL

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Local

My Hub

59.23 GB / 30.43 GB in use

7 Images

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☐	Name	Tag	Image ID	Created	Size	Actions
☐	nvr.io/nvidia/pytorch	26.01-py3	38ed2ecb2c16	1 month ago	30.43 GB	👤 🔍 📄 🗑️

Last refresh: 1 day ago

Terminal

```
PS C:\Users\afeef> docker run nvr.io/nvidia/pytorch:26.01-py3

=====
== PyTorch ==
=====

NVIDIA Release 26.01 (build 256811083)
PyTorch Version 2.10.0a0+436e1d3
Container Image Copyright (c) 2025, NVIDIA CORPORATION & AFFILIATES. All rights reserved.
Copyright (c) 2014-2024 Facebook Inc.
Copyright (c) 2011-2014 Idiap Research Institute (Roman Collobert)
Copyright (c) 2012-2014 DeepMind Technologies (Koray Kavukcuoglu)
Copyright (c) 2011-2012 NEC Laboratories America (Koray Kavukcuoglu)
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Copyright (c) 2015 Google Inc.
Copyright (c) 2015 Yangqing Jia
Copyright (c) 2013-2016 The Caffe contributors
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Engine running

RAM 1.54 GB CPU 0.00% Disk: 89.19 GB used (limit 1006.85 GB)

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Figure 11: PyTorch Run

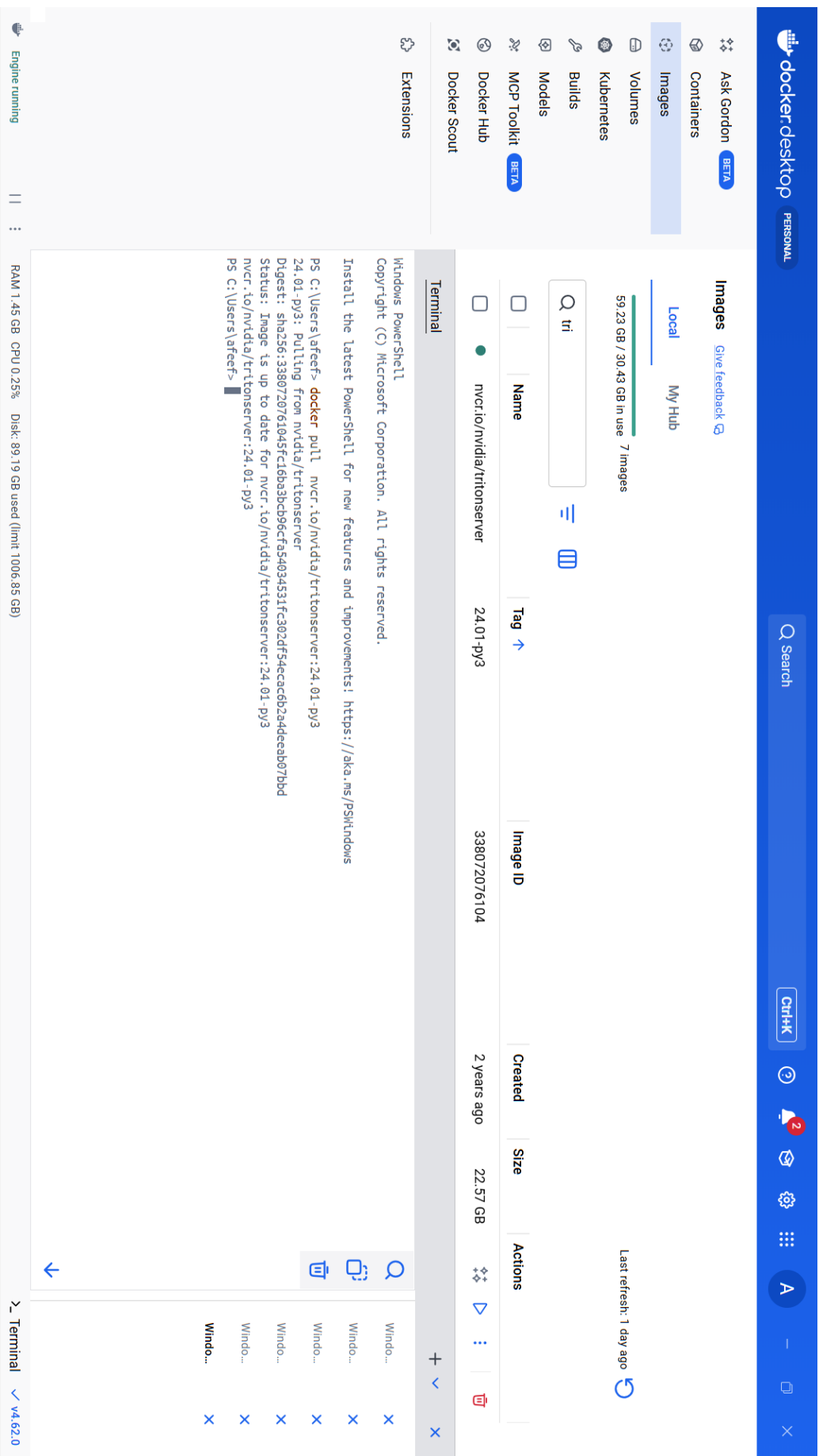


Figure 12: Use Case Container Pull

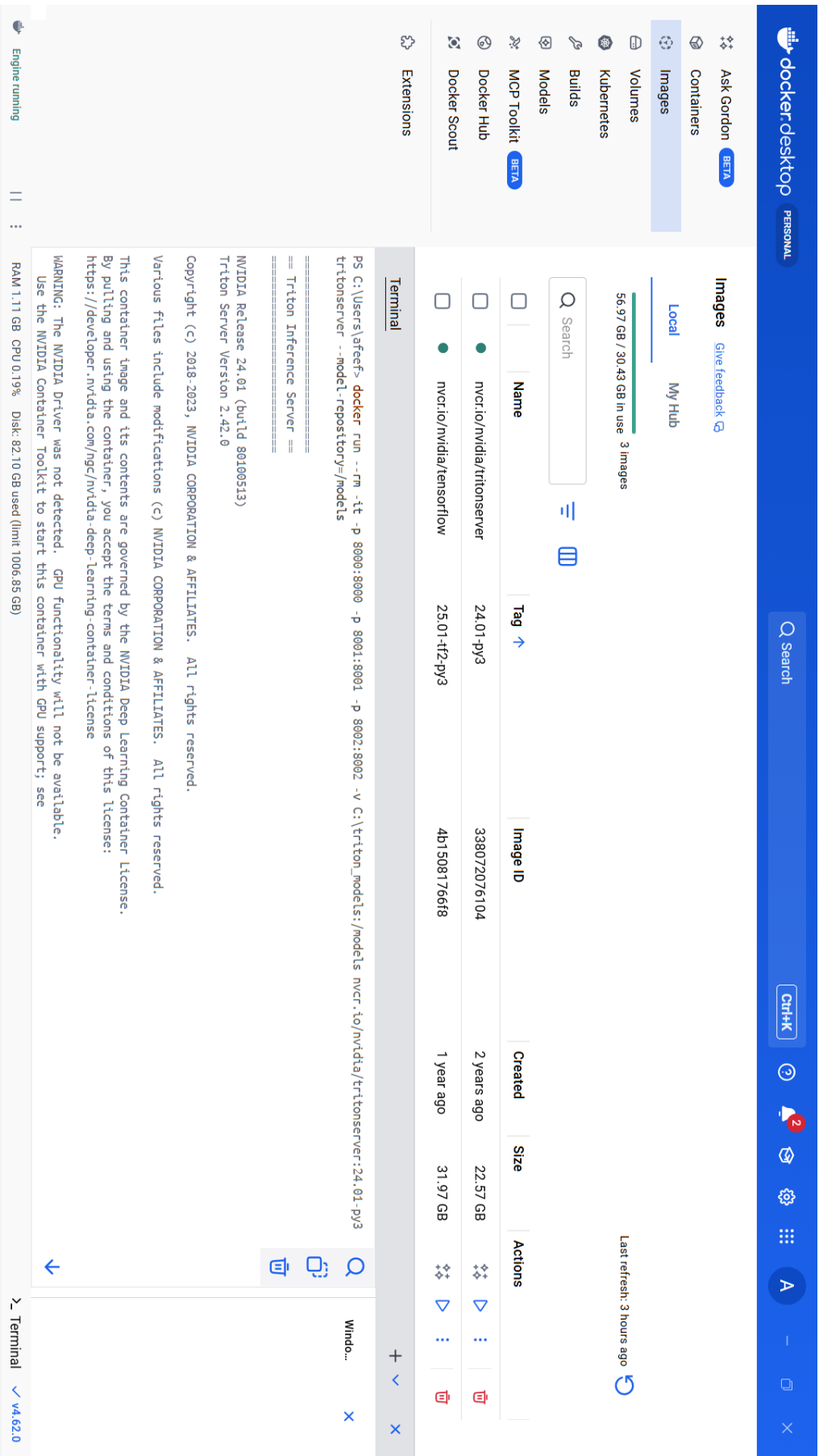


Figure 13: Use Case Container Run

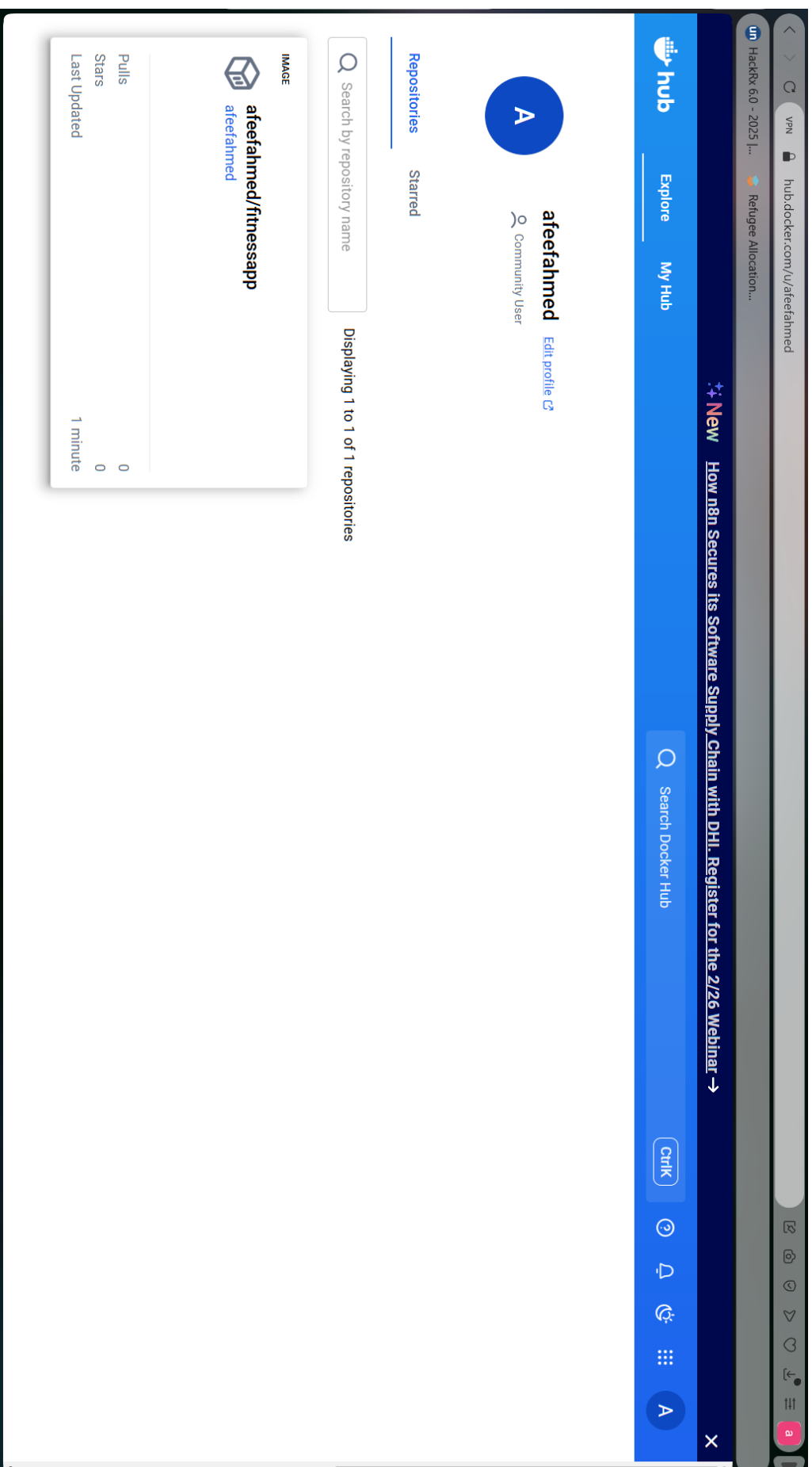


Figure 14: Self Project Pushed in Docker Hub

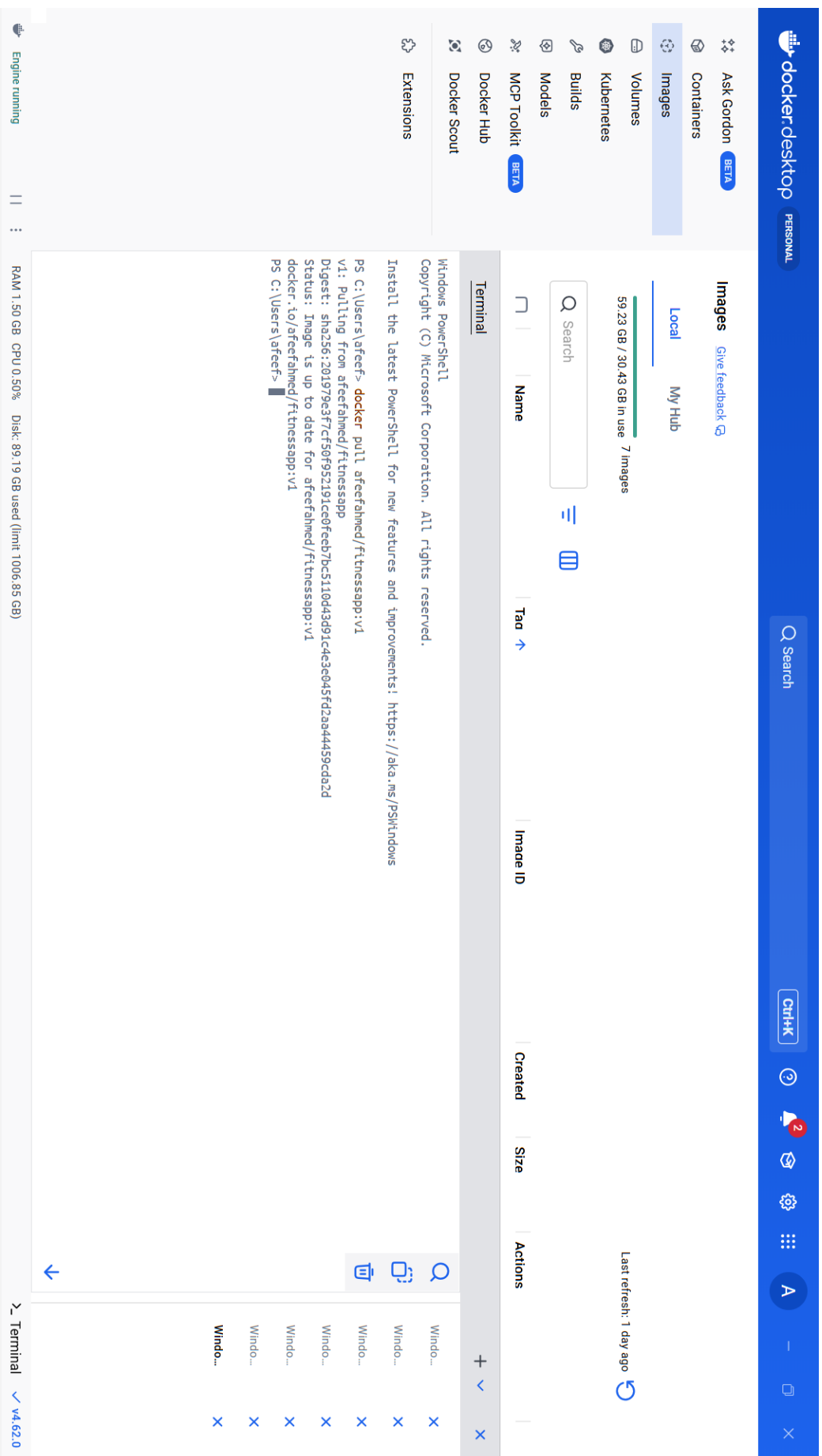


Figure 15: Pulling Self Image

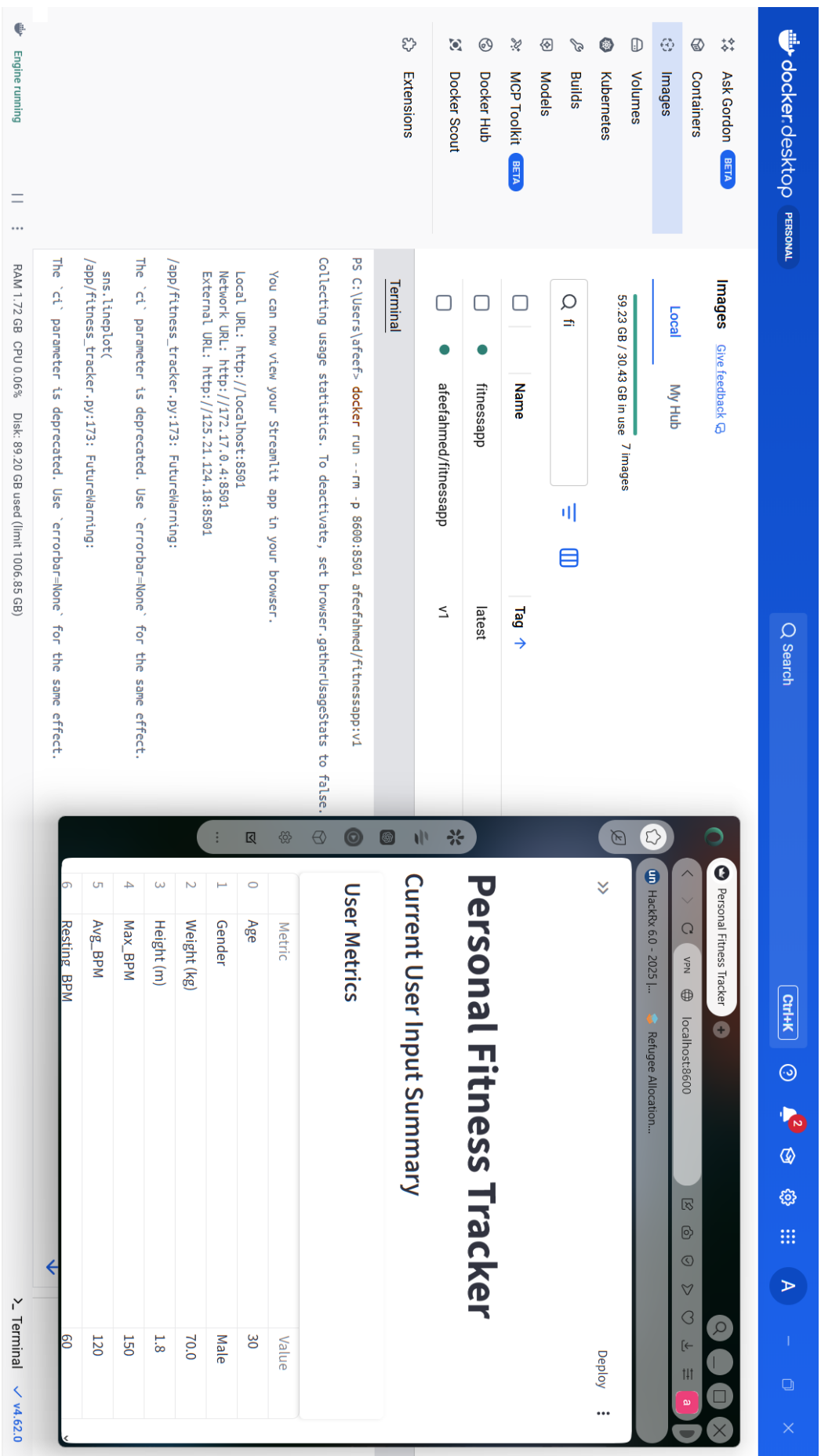


Figure 16: Running self-image

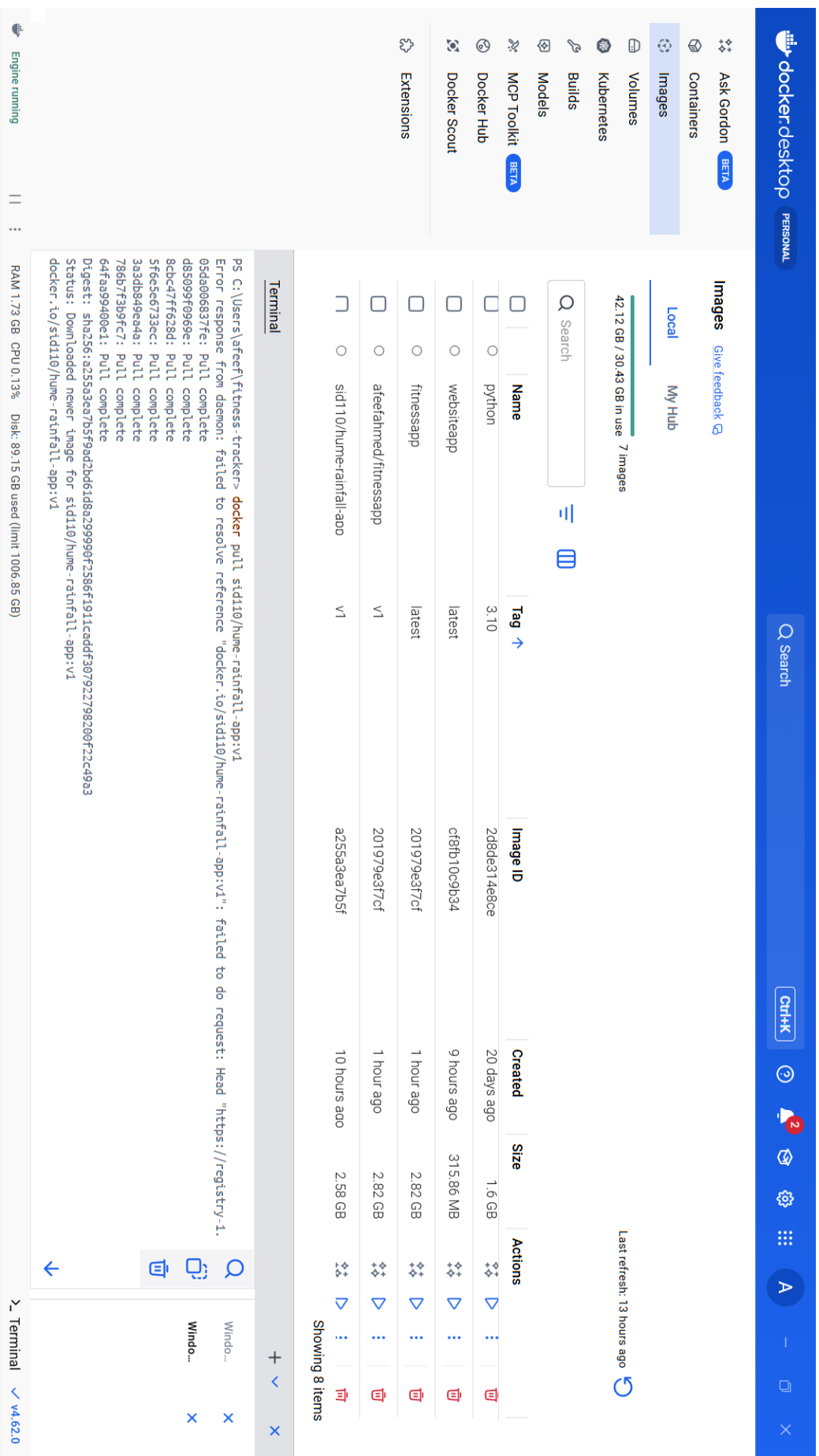


Figure 17: Pulling Friend Image

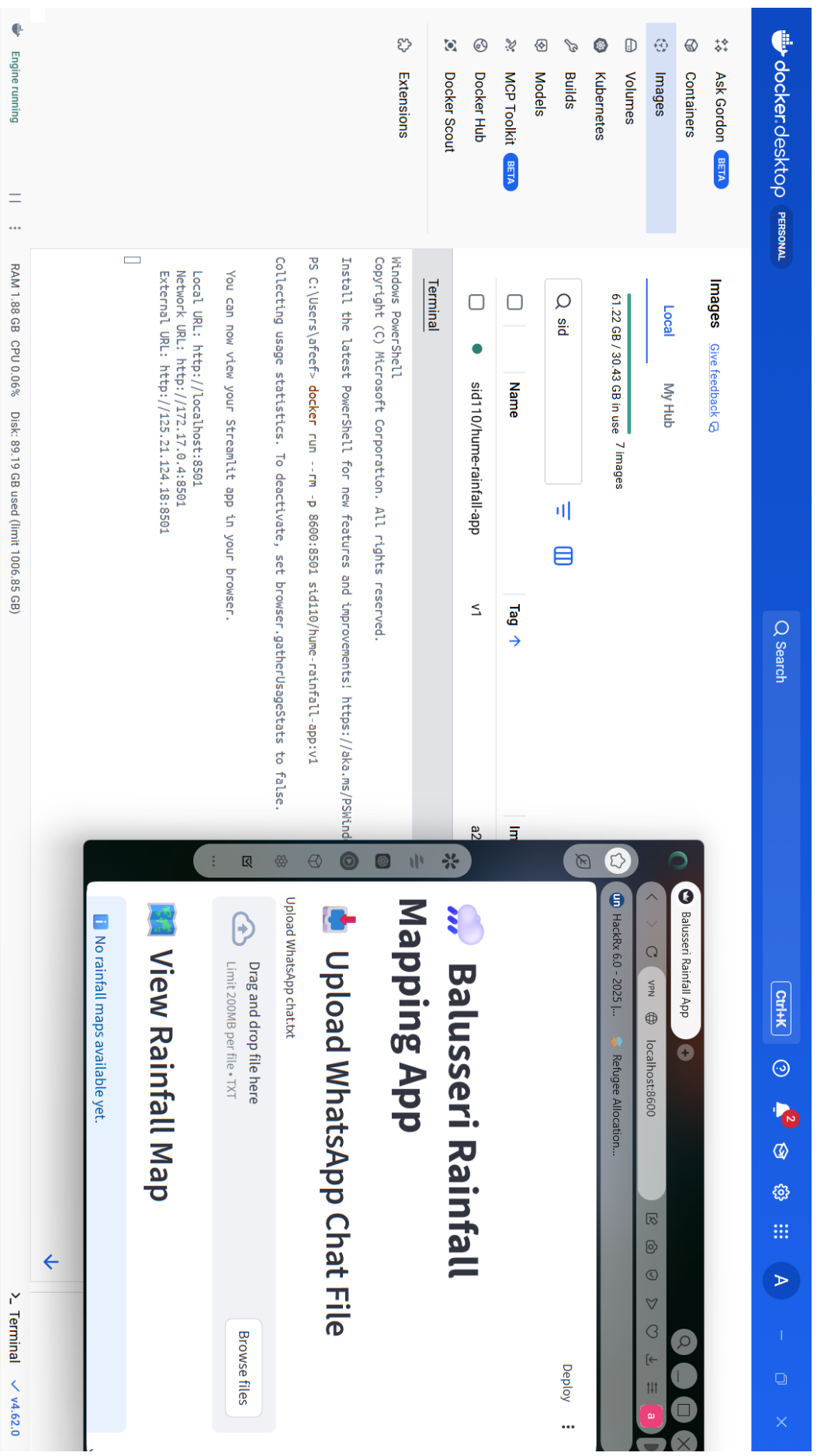


Figure 18: Running Friend Image

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dockerdesktop

PERSONAL

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Created

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Actions

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Terminal

Windows PowerShell

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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\afreef> kubectl get nodes

NAME STATUS ROLES AGE VERSION

minikube Ready control-plane 22h v1.35.0

PS C:\Users\afreef> kubectl get pods -A

NAMESPACE NAME READY STATUS RESTARTS AGE

default fitness-deployment-84d6b4b84-1668l 1/1 Running 0 12m

kube-system fitness-secret-deployment-5cc7dc7fd4-6lrj5 1/1 Running 0 12m

kube-system coredns-7d764666f9-s1xbz 1/1 Running 0 12m

kube-system coredns-7d764666f9-z4gpf 1/1 Running 0 12m

kube-system etcd-minikube 1/1 Running 0 22h

kube-system kube-apiserver-minikube 1/1 Running 0 22h

kube-system kube-controller-manager-minikube 1/1 Running 1 (8h ago) 22h

kube-system kube-proxy-mj9n 1/1 Running 0 22h

kube-system kube-scheduler-minikube 1/1 Running 0 22h

PS C:\Users\afreef>

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Engine running

RAM 1.62 GB CPU 0.00% Disk: 89.19 GB used (limit 1006.85 GB)

Terminal v4.62.0

Figure 20: Checking Nodes & Pods After Installing

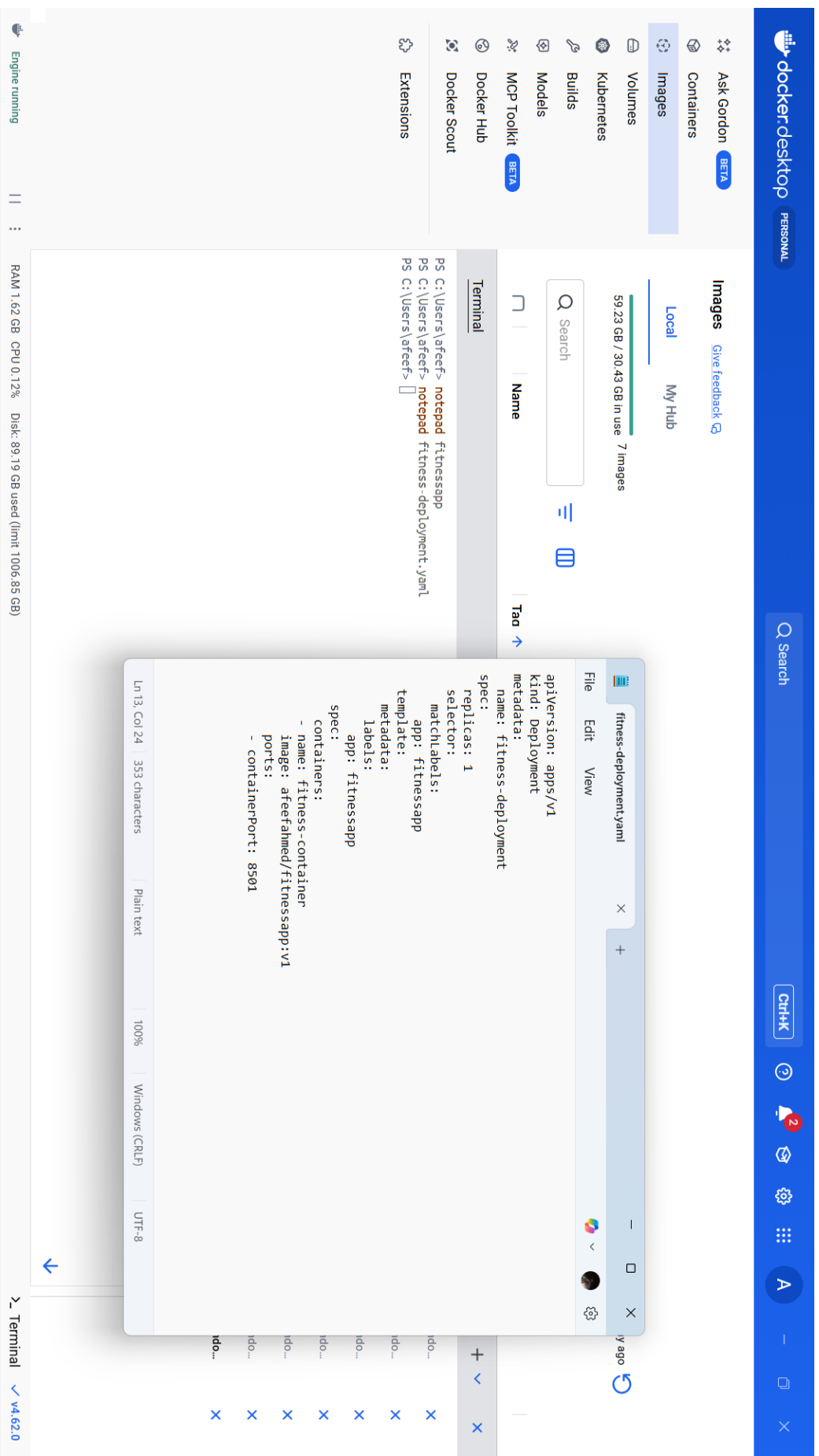


Figure 21: Yaml Defining For Self Pull

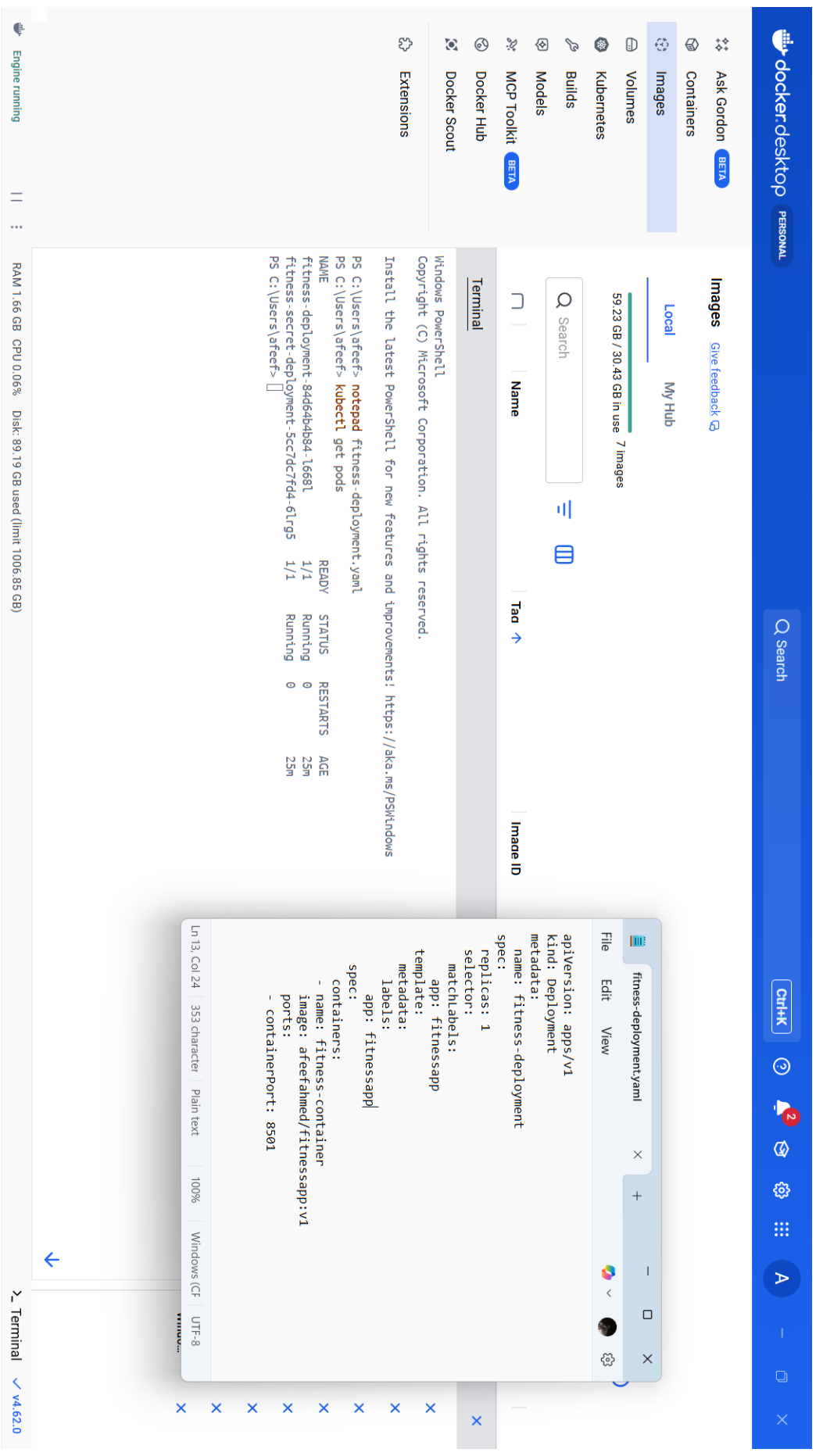


Figure22: SelfPullExecution

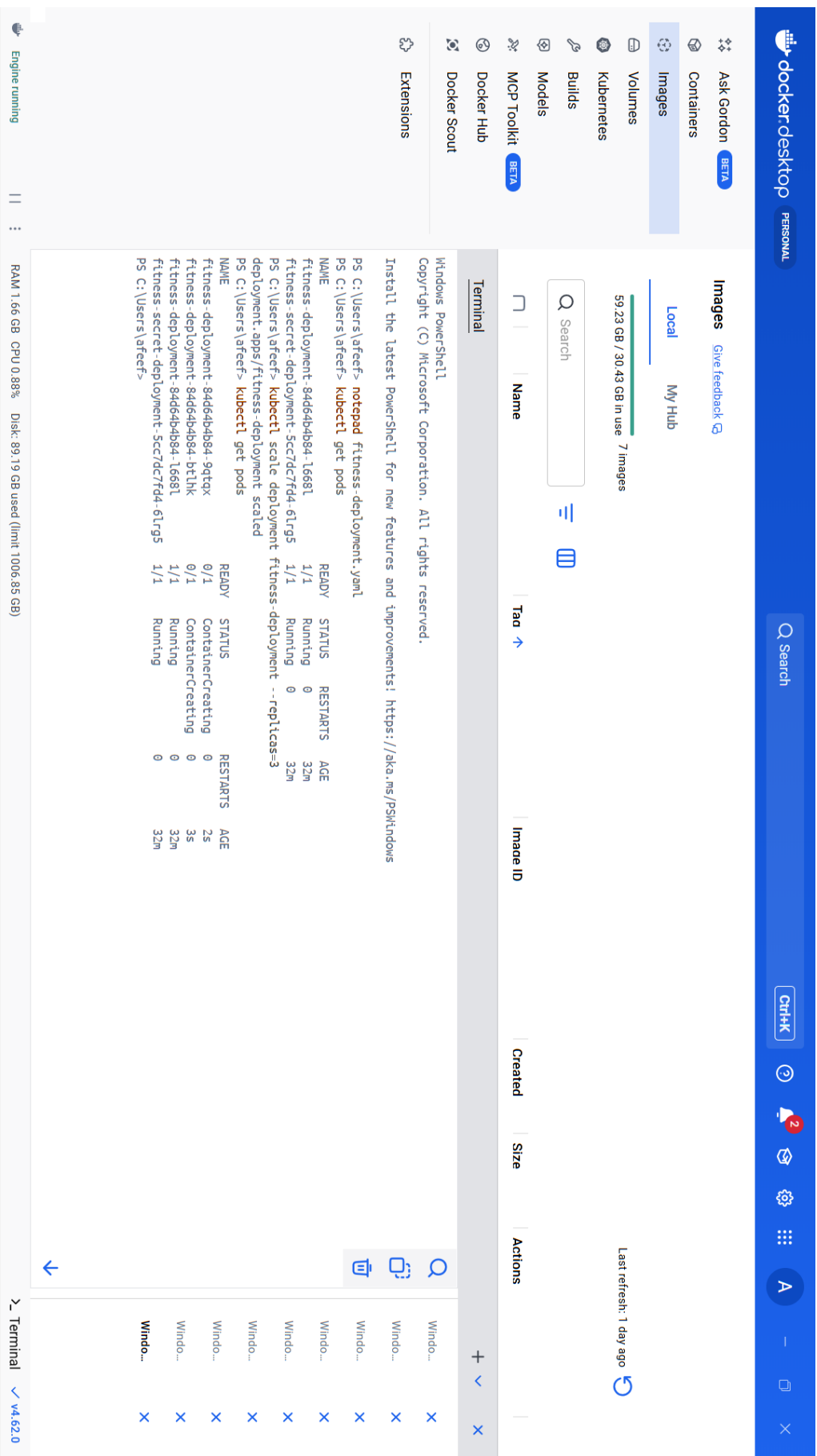


Figure 23: UPSCALE

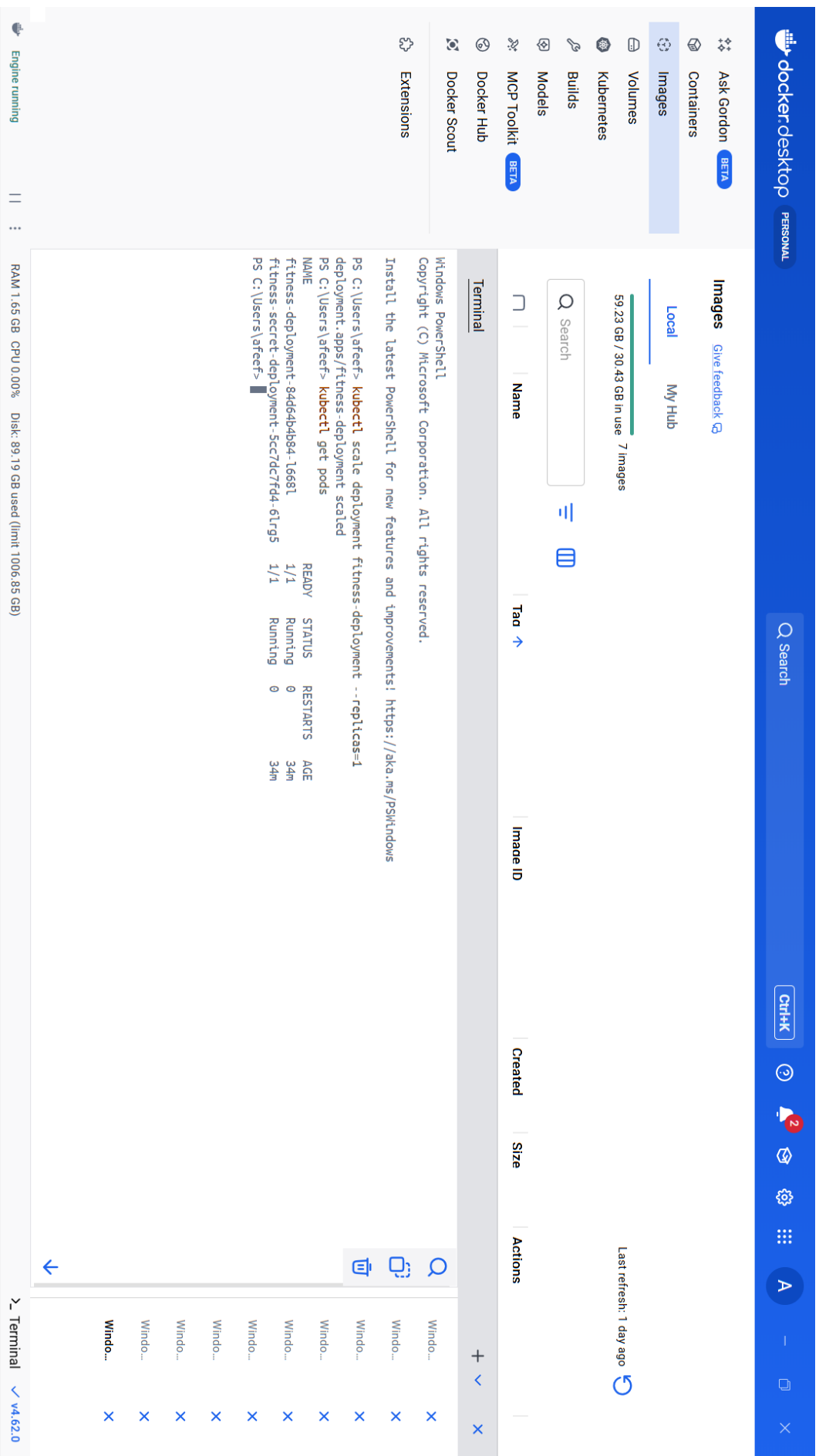


Figure 24: Downscale

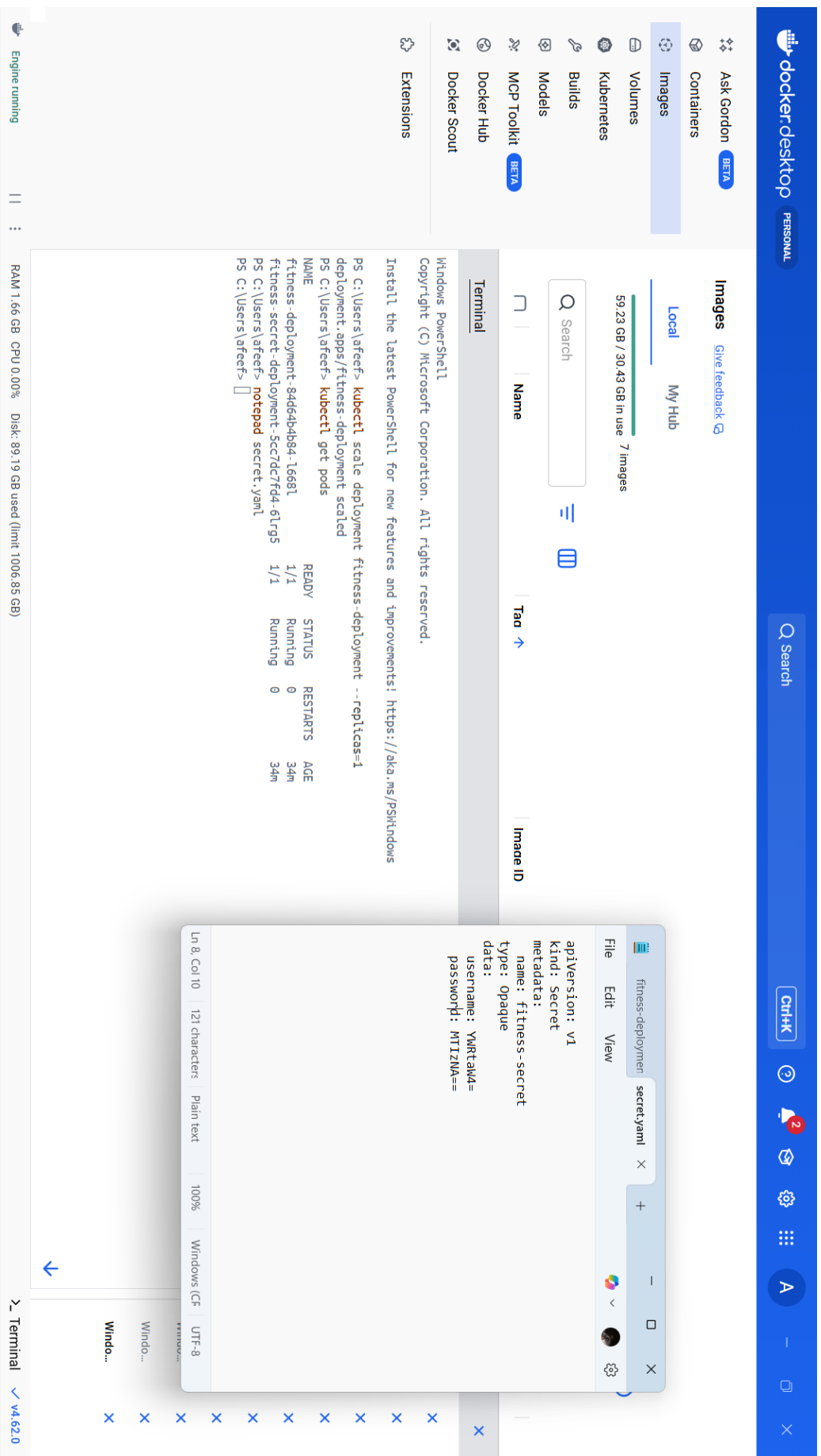


Figure25: Secret Yaml Defining

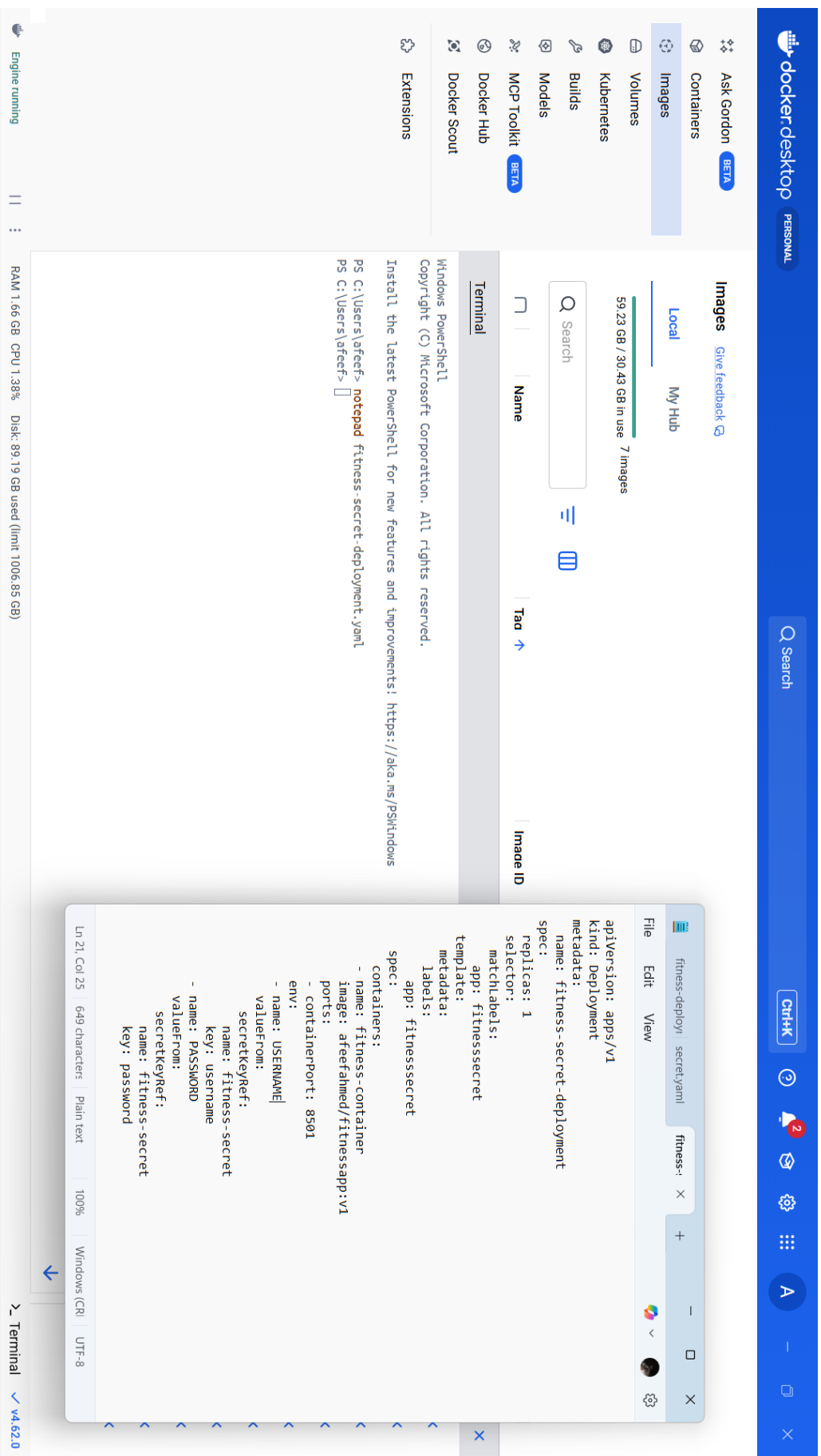


Figure 26: New Deployment

dockerdesktop

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Tag

Image ID

Created

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Terminal

Windows PowerShell

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PS C:\Users\jafeef> notepad fitness-secret-deployment.yaml

PS C:\Users\jafeef> kubectl create secret generic app-secret --from-literal=APP_MODE=production

secret/app-secret created

PS C:\Users\jafeef> kubectl get secrets

NAME TYPE DATA AGE

app-secret Opaque 1 5s

fitness-secret Opaque 2 23h

PS C:\Users\jafeef> kubectl get secret app-secret -o yaml

apiVersion: v1

data:

APP_MODE: CHJvZHVjdGlvbg==

kind: Secret

metadata:

creationTimestamp: "2026-02-26T08:07:22Z"

name: app-secret

namespace: default

resourceVersion: "9356"

uid: e7354792-8040-4195-b23a-a892507230cf

type: Opaque

PS C:\Users\jafeef> kubectl get secret app-secret -o jsonpath="{.data.APP_MODE}" | %([System.Text.Encoding]::UTF8.GetString([System.Convert]::Fro

RAM 1.66 GB CPU 1.50% Disk: 89.19 GB used (limit 1006.85 GB)

Engine running

Terminal

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Figure27: GettingSecretFor Check

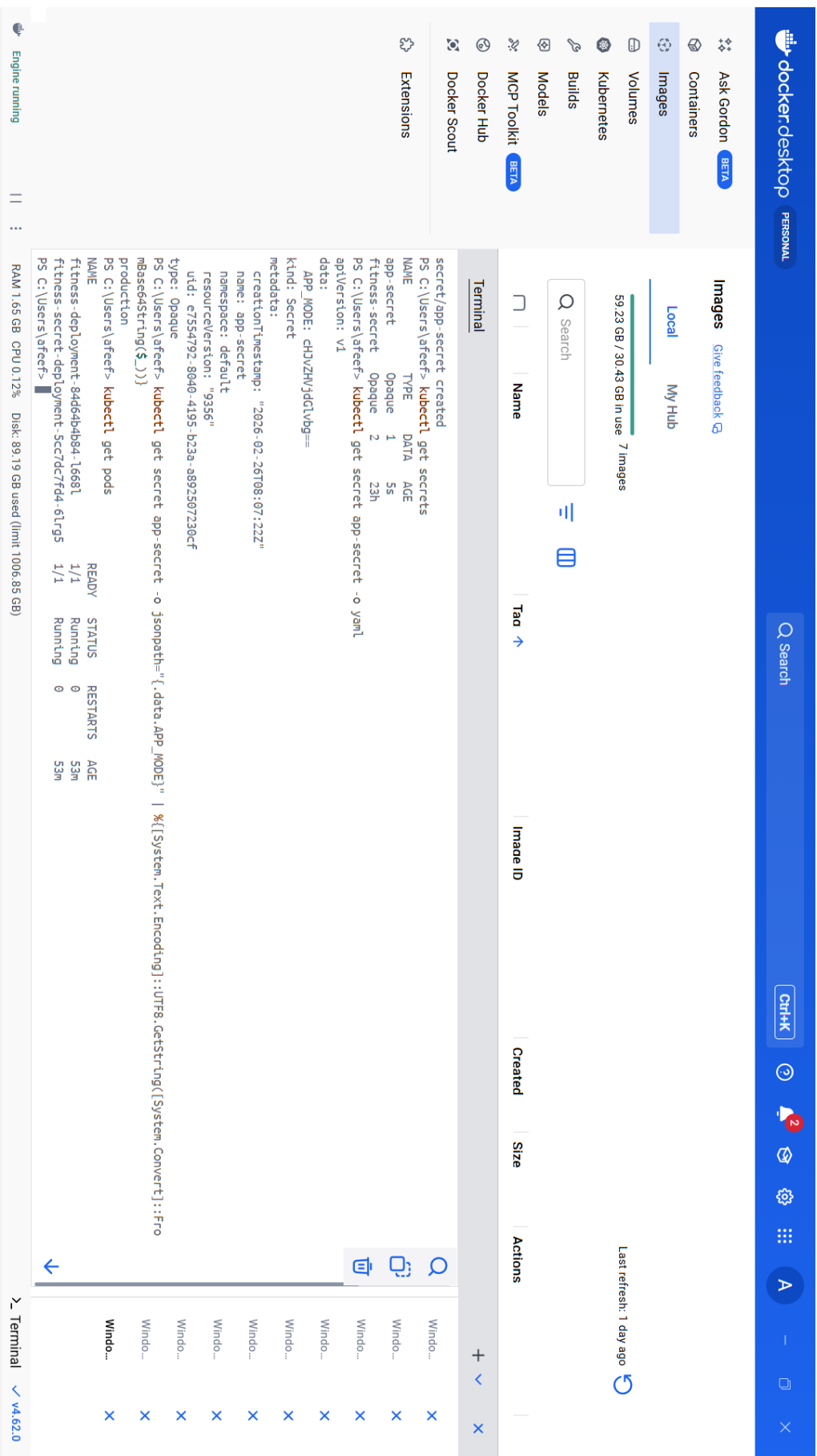


Figure 28: Secret Deployment

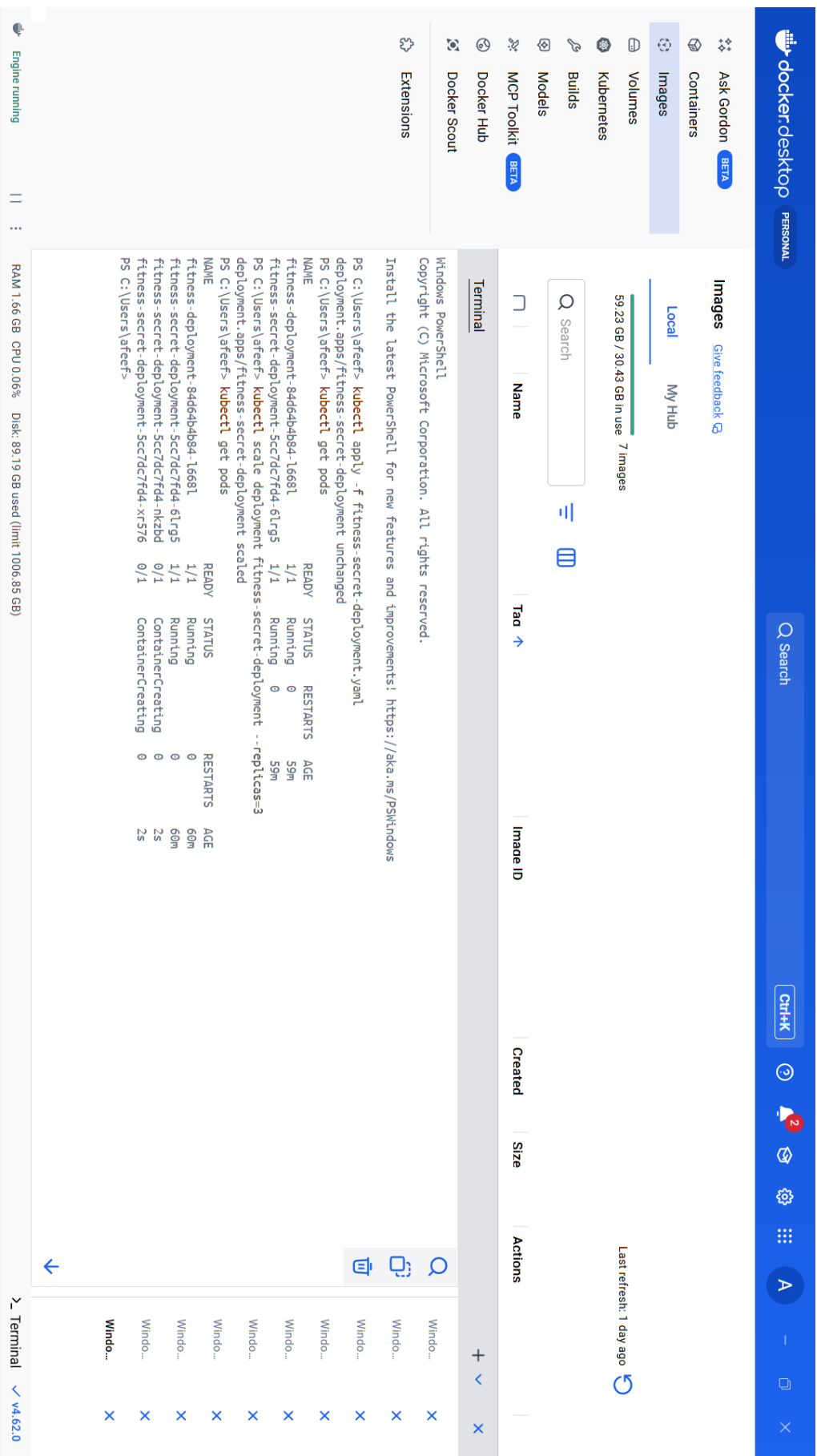


Figure 29: UPSCALE

dockerdesktop

PERSONAL

Search

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7 Images

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Terminal

PS C:\Users\jafeef> kubectl get pods

NAME	STATUS	RESTARTS	AGE	
fitness-deployment-84d64b4b94-16681	1/1	Running	0	59m
fitness-secret-deployment-5cc7dc7fd4-61rng5	1/1	Running	0	59m
PS C:\Users\jafeef> kubectl scale deployment fitness-secret-deployment --replicas=3				
deployment.apps/fitness-secret-deployment scaled				
PS C:\Users\jafeef> kubectl get pods				
NAME	STATUS	RESTARTS	AGE	
fitness-deployment-84d64b4b94-16681	1/1	Running	0	60m
fitness-secret-deployment-5cc7dc7fd4-61rng5	1/1	Running	0	60m
fitness-secret-deployment-5cc7dc7fd4-nkzbd	0/1	ContainerCreating	0	2s
fitness-secret-deployment-5cc7dc7fd4-xr576	0/1	ContainerCreating	0	2s
PS C:\Users\jafeef> kubectl scale deployment fitness-secret-deployment --replicas=1				
deployment.apps/fitness-secret-deployment scaled				
PS C:\Users\jafeef> kubectl get pods				
NAME	STATUS	RESTARTS	AGE	
fitness-deployment-84d64b4b94-16681	1/1	Running	0	63m
fitness-secret-deployment-5cc7dc7fd4-61rng5	1/1	Running	0	63m
fitness-secret-deployment-5cc7dc7fd4-nkzbd	1/1	Terminating	0	3m32s
fitness-secret-deployment-5cc7dc7fd4-xr576	1/1	Terminating	0	3m32s
PS C:\Users\jafeef> kubectl get pods				
NAME	STATUS	RESTARTS	AGE	
fitness-deployment-84d64b4b94-16681	1/1	Running	0	63m
fitness-secret-deployment-5cc7dc7fd4-61rng5	1/1	Running	0	63m
fitness-secret-deployment-5cc7dc7fd4-nkzbd	1/1	Running	0	63m
PS C:\Users\jafeef>				

RAM 1.66 GB CPU 0.06% Disk: 89.19 GB used (limit 1006.85 GB)

Terminal

v4.62.0

Figure30: Downscaling

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7 images

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Q Search

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Q Search

Name

Tag

Image ID

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PS C:\Users\afeef> kubectl get deployments

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
fitness-deployment	1/1	1	1	23h
fitness-secret-deployment	1/1	1	1	23h
hume-rainfall-deployment	1/1	1	1	2m59s

PS C:\Users\afeef> kubectl describe deployment fitness-secret-deployment

NAME: fitness-secret-deployment
Namespace: default
CreationTimestamp: Wed, 25 Feb 2026 14:22:42 +0530
Labels: <none>
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=fitnesssecret
Replicas: 1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
Labels: app=fitnesssecret
Containers:
fitness-container:
Image: afeefahmed/fitnessapp:v1
Port: 8501/TCP
Host Port: 0/TCP
Environment:
USERNAME: <set to the key 'username' in secret 'fitness-secret'> Optional: false

RAM 1.55 GB CPU 1.75% Disk 89.19 GB used (limit 1006.85 GB)

> Terminal

✓ v4.62.0

Figure31: AllFinal Deployment

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MCP Toolkit BETA

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Image ID

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Size

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PS C:\Users\afeef> kubectl get deployments

NAME	READY	UP-T-O-DATE	AVAILABLE	AGE
fitness-deployment	1/1	1	1	23h
fitness-secret-deployment	1/1	1	1	23h
hume-rainfall-deployment	1/1	1	1	4m44s

PS C:\Users\afeef>

Terminal

Windows PowerShell

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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\afeef> kubectl get deployments

NAME	READY	UP-T-O-DATE	AVAILABLE	AGE
fitness-deployment	1/1	1	1	23h
fitness-secret-deployment	1/1	1	1	23h
hume-rainfall-deployment	1/1	1	1	4m44s

PS C:\Users\afeef>

RAM 1.55 GB

CPU 1.00%

Disk: 89.19 GB used (limit 1006.85 GB)

Search

Ctrl+K

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Last refresh: 2 days ago

Figure 32: All Pods